

ENGLISH

-WhitePEAK

9) You are Arun Sharma, a resident of plot no. 42, Nehru Avenue, Ranchi. Write a letter to the Editor of the Ranchi Times, highlighting the growing problem of Water shortage in your locality. Suggest some practical measures that the public and the authorities can take to conserve water.

MATH

25/11/25

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- a) The length and breadth of the rect. field are in the ratio 3:2. If the area of the field is 3456 m^2 . If the cost of fencing is ₹ 60/m.

Let the length of the field be $(3u) \text{ m}$ and breadth be $(2u) \text{ m}$.

Area of the rect. field is given as 3456 m^2

$$\begin{aligned} 3u \times 2u &= 3456 \\ \Rightarrow 6u^2 &= 3456 \\ \Rightarrow u^2 &= \frac{3456}{6} = 576 \\ &= 24^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow u^2 &= 576 \\ \Rightarrow u^2 &= 24^2 \end{aligned}$$

$$\therefore \boxed{u = 24}$$

Now,

$$\begin{aligned} \text{length of the field} &= (3 \times 24) \text{ m} = 72 \text{ m} \\ \& \text{ breadth} &= (2 \times 24) = 48 \text{ m} \\ \text{Perimeter of the field} &= 2(l+b) \\ &= 2(72 + 48) \text{ m} \end{aligned}$$

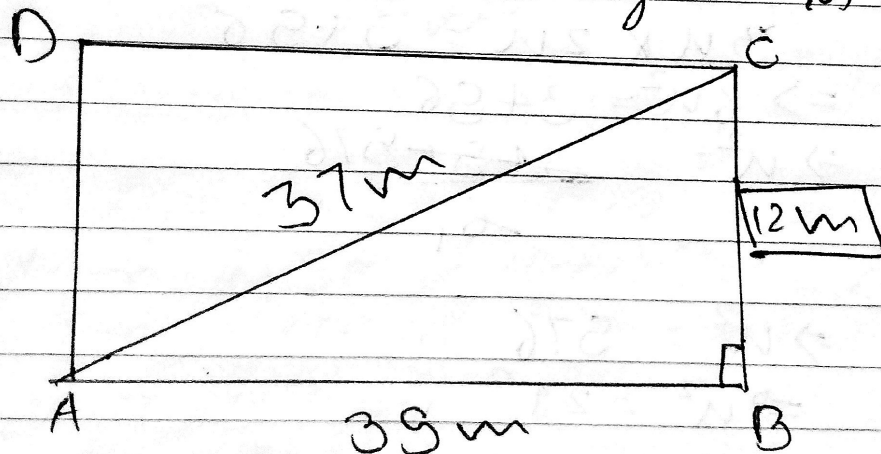
$$= 2 \times 120 \text{ m}$$

$$= 240 \text{ m}$$

$$\therefore \text{Cost of fencing} = ₹ 60 \text{ m}$$

$$\therefore \text{Cost of fencing } 240 \text{ m} = ₹ (60 \times 240) \\ = ₹ 14400$$

Q Find the area of a rect. plot, one side which measures 35m and the diagonal is 37m.



In right ΔABC ;

$$AC^2 = AB^2 + BC^2 \text{ [Pythagoras Theorem]}$$

$$\Rightarrow (37)^2 = (35)^2 + (BC)^2$$

$$\Rightarrow (37)^2 - (35)^2 = (BC)^2$$

$$\Rightarrow (37+35)(37-35) = BC^2$$

$$\Rightarrow 72 \times 2 = BC^2$$

$$\Rightarrow 144 = BC^2$$

$$\Rightarrow 12^2 = BC^2$$

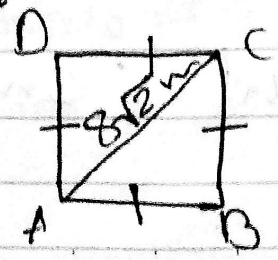
$$\therefore \underline{BC = 12}$$

So, Breadth of the field is = 12m

$$\begin{aligned} \therefore \text{Area of the field} &= 2(l+b) \\ &= 2(35+12) \\ &= 2 \times 47 \\ &= 94 \text{ m} \end{aligned}$$

$$\begin{aligned} &= (l \times b) \\ &= (35 \times 12) \text{ m}^2 \\ &= 420 \text{ m}^2 \end{aligned}$$

Q) Find the area of a sq. the length of whose diagonal is $8\sqrt{2}$ m.



Diagonal of the sq. = $8\sqrt{2}$ m.

$$\text{Area of a sq.} = \frac{1}{2} \times (\text{diagonal})^2$$

$$= \frac{1}{2} \times (8\sqrt{2})^2 \text{ m}^2$$

$$= \frac{1 \times 8\sqrt{2} \times 8\sqrt{2}}{2} \text{ m}^2$$

$$= \frac{1}{2} \times 64 \times 2 \text{ m}^2$$

$$= 64 \text{ m}^2$$

★) A room is 9m long, 8m broad and 6.5m high. It has one door of dimensions (2m x 1.5m) and four windows each of dimensions (1.5m x 1m). Find the cost of white washing the walls at ₹ 25/m².

length = 9m,
breadth = 8m, height = 6.5m

Area of 4 walls of the room

$$\begin{aligned}
 &= 2(l+b) \times h \\
 &= 2(9+8) \times 6.5 \text{ m}^2 \\
 &= 2 \times 17 \times 6.5 \text{ m}^2 \\
 &= \cancel{2} \times 17 \times \cancel{6.5}^{13} \text{ m}^2 \\
 &\quad \quad \quad \downarrow \quad \downarrow \\
 &\quad \quad \quad 10 \quad 5 \\
 &= 221 \text{ m}^2
 \end{aligned}$$

Now,

$$\begin{aligned}
 \text{Area of 1 door} &= l \times b \\
 &= 1.5 \times 2 \text{ m}^2 \\
 &= 3 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of 4 windows} &= \left(\frac{3}{2} \times 1 \times 4 \right) \text{ m}^2 \\
 &= 6 \text{ m}^2
 \end{aligned}$$

Total area of doors and windows

$$= (3+6) \text{ m}^2 = 9 \text{ m}^2$$

$$\begin{aligned}
 \text{Area to be white washed} &= (221-9) \text{ m}^2 \\
 &= 212 \text{ m}^2
 \end{aligned}$$

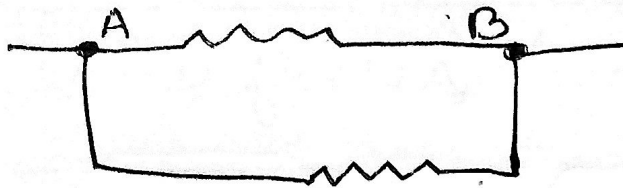
$$\begin{aligned}
 \text{Rate of whitewashing} &= ₹ 25/\text{m}^2 \\
 \text{Cost of whitewashing} &= ₹ (212 \times 25) \\
 &= \boxed{₹ 5300}
 \end{aligned}$$

PHYSICS

29/11/21

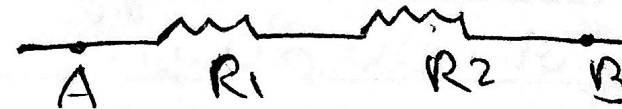
Resistance

Parallel



Two path joining
at one place (A, B)

Series



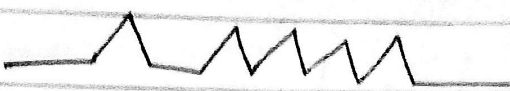
Two resistance is
attached with series

Equivalent = $R_1 + R_2$

Resistance $\rightarrow$ equivalent
resistance

$$\frac{1}{R_{\text{equivalent}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Resistance - A property of matter which opposes or resist the movement of current.

Symbol: 

Unit of resistance ohm (Ω)

Ohm's law

In conductor Potential difference is directly proportional to electric current.

$$V \propto I$$

Potential difference - amt. of work done to bring unit positive charge from infinity to that point.
Volt.

Current - Rate of flow of charge through conductor Ampere (Unit)